CS 2840 Fall 2025
11/04/2025

Scribes: Kumar Tanmay, Aditya Saxena
Based on notes by: Kumar Tanmay, Aditya Saxena

Lecture 17:Distributional Preference Alignment of LLMs via Optimal Transport

1 Introduction and Motivation

Large Language Models (LLMs) are typically aligned with human preferences by maximizing an expected reward under a proximity constraint to a reference model. The goal is to find a policy ω_ε solving:

$$\max_{\omega} \ \mathbb{E}_\omega[r(x,y)] \quad \text{s.t.} \quad \mathrm{KL}(\omega \,\|\, \omega_{\text{ref}}) \le \varepsilon.$$

The optimal solution takes the Gibbs form:

$$\omega_\varepsilon(y|x) = \omega_{\text{ref}}(y|x)\frac{\exp(r(x,y)/\varepsilon)}{Z(x)}.$$

This forms the theoretical foundation for Direct Preference Optimisation (DPO) [RSM+23], where the model learns from pairwise preference data (x, y^+, y^-), a chosen and rejected response.

2 From Pointwise to Distributional Preference

DPO minimizes a logistic loss:

$$L_{\text{DPO}} = \mathbb{E}_{(x,y^+,y^-)}\Big[-\log\sigma\big(\beta(\log\tfrac{\omega(y^+|x)}{\omega_{\text{ref}}(y^+|x)} - \log\tfrac{\omega(y^-|x)}{\omega_{\text{ref}}(y^-|x)})\big)\Big],$$

which enforces pointwise preference:

$$\log\tfrac{\omega(y^+|x)}{\omega_{\text{ref}}(y^+|x)} > \log\tfrac{\omega(y^-|x)}{\omega_{\text{ref}}(y^-|x)}.$$

While effective, DPO only guarantees sample-level ordering and struggles in unpaired or distributional settings.

2.1 First-Order Stochastic Dominance (FSD)

The lecture introduces a distributional notion of preference: First-Order Stochastic Dominance (FSD). For random variables Z_1, Z_2 with quantile functions Q_{Z_1} and Q_{Z_2}:

$$Z_1 \succeq_{\text{FSD}} Z_2 \iff Q_{Z_1}(p) \ge Q_{Z_2}(p), \quad \forall p \in [0,1].$$

Intuitively, Z_1 dominates Z_2 if at every percentile, its reward distribution yields higher values. In the context of alignment, this means the distribution of rewards for "good" responses should dominate that for "bad" ones.

3 Distributional Preference Alignment [MMB+24]

3.1 Unpaired Setting

We have two independent datasets:

- μ^+: Distribution of positive (chosen) (X^+, Y^+) pairs.
- μ^-: Distribution of negative (rejected) (X^-, Y^-) pairs.

We require:

$$\log\tfrac{\omega_\varepsilon(Y^+|X^+)}{\omega_{\text{ref}}(Y^+|X^+)} \succeq_{\text{FSD}} \log\tfrac{\omega_\varepsilon(Y^-|X^-)}{\omega_{\text{ref}}(Y^-|X^-)}.$$

3.2 Paired Setting

Given triplets (x, y^+, y^-):

$$\log \frac{\omega_\varepsilon(y^+|x)}{\omega_\varepsilon(y^-|x)} \succeq_{\text{FSD}} \log \frac{\omega_{\text{ref}}(y^+|x)}{\omega_{\text{ref}}(y^-|x)}.$$

Thus, the log-likelihood margin for positive vs. negative responses under the model should stochastically dominate the same margin under the reference model.

4 Relaxation via Optimal Transport

FSD defines a non-convex constraint. The lecture introduces a convex relaxation minimizing its violation:

$$\min_\theta \int_0^1 h(Q_U(t) - Q_V(t))\, dt,$$

where h penalizes negative margins. Examples:

$$h(x) = \begin{cases} (\beta - x)_+^2 & \text{Hinge-squared,} \\ \log(1 + \exp(-\beta x)) & \text{Logistic,} \\ (\beta - x)^2 & \text{Least Squares.} \end{cases}$$

Using results from *Santambrogio (2015)* [**?**],

$$\int_0^1 h(Q_U(t) - Q_V(t))\, dt = \min_{\gamma \in \Pi(\mu_U, \mu_V)} \int h(u - v)\, d\gamma(u, v),$$

showing that the relaxed objective equals a one-dimensional optimal transport (OT) cost.

4.1 Empirical Computation

For empirical measures $\hat{\mu}_U = \frac{1}{n}\sum_i \delta_{u_i}$ and $\hat{\mu}_V = \frac{1}{n}\sum_i \delta_{v_i}$,

$$\text{OT}_h(\hat{\mu}_U, \hat{\mu}_V) = \frac{1}{n}\sum_{i=1}^n h(u_{(i)} - v_{(i)}),$$

where $(u_{(i)})$, $(v_{(i)})$ are sorted values. This leads to Alignment via Optimal Transport (AOT):

$$\min_\theta \text{OT}_h((r_\theta \circ \pi_\theta)_\# \mu^+, (r_\theta \circ \pi_\theta)_\# \mu^-).$$

5 AOT Algorithms

Two versions are defined:

Algorithm 1: AOT-Unpaired.

1. Sample minibatches from μ^+ and μ^-.
2. Compute rewards u_i, v_i as log-likelihood ratios vs. reference.
3. Sort u_i, v_i and compute loss $\frac{1}{b}\sum_i \log \sigma(\beta(u_{(i)} - v_{(i)}))$.
4. Update θ with AdamW.

Algorithm 2: AOT-Paired. Uses margins $\log \frac{\omega(y^+|x)}{\omega(y^-|x)}$ in place of raw rewards.

6 Theoretical Guarantees

Under boundedness assumptions, AOT generalizes at the *parametric rate*:

$$\mathbb{E}\left[\mathrm{OT}_h(\hat{\mu}_U, \hat{\mu}_V) - \mathrm{OT}_h(\mu_U, \mu_V)\right] = O(n^{-1/2}).$$

This arises from a dual OT formulation with c-concave potentials and Rademacher complexity bounds.

7 Empirical Results

7.1 Quantile Dominance

Figure 1 (Slide 28) shows quantile plots where AOT's reward distribution for chosen samples dominates rejected ones, validating FSD empirically. Competing methods (DPO, KTO, IPO) display smaller gaps.

7.2 Benchmark Comparison

Experiments on the Merlinite-7B model trained with UltraFeedback show AOT outperforming all baselines on AlpacaEval (GPT-4) and remaining competitive across ARC, HellaSwag, MMLU, TruthfulQA, Winogrande, and GSM8K.

7.3 Ablation Studies

Logistic loss performs best among penalty types, and larger batch sizes improve dominance. Smaller β values (e.g. 0.01) enable stronger alignment by allowing more policy flexibility.

7.4 Toxicity Evaluation

Using synthetic UltraChat prompts and AttaQ dataset scored via Perspective API, AOT-trained models show reduced toxicity across categories like *insult*, *threat*, and *identity attack*.

8 Multivariate Stochastic Dominance via Optimal Transport

The latter part of the lecture introduced the work of Rioux et al. (NeurIPS 2024), which generalizes the notion of first-order stochastic dominance to multivariate settings. While AOT enforces dominance between one-dimensional reward distributions, multivariate stochastic dominance (MSD) extends this concept to vector-valued outcomes, enabling joint comparison of multiple performance criteria such as helpfulness, honesty, and safety.

Formally, for d-dimensional random vectors X, Y, MSD defines $X \succeq_{\mathrm{FSD}} Y$ if the optimal transport cost between their distributions μ and ν under a separable convex cost $c(x, y) = \sum_{i=1}^{d} h(y_i - x_i)$ equals zero. To improve both computational and statistical efficiency, the method introduces an *entropically regularized* OT objective:

$$\mathrm{OT}_{c,\eta}(\mu, \nu) = \inf_{\gamma \in \Pi(\mu,\nu)} \left[\int c(x, y)\, d\gamma(x, y) + \eta\, \mathrm{KL}(\gamma \,\|\, \mu \otimes \nu) \right],$$

and defines a normalized dominance score $\varphi_{h,\eta}(\mu, \nu) \in [0, 1]$ that quantifies directional preference between distributions.

Empirically, MSD was used to benchmark 12 large language models based on multiple automated metrics, producing rankings with higher Kendall–τ correlation to human (ChatGPT) evaluations than scalar or copula-based aggregation methods. This demonstrates how multivariate OT provides a robust foundation for fair and interpretable model comparison.

9 Conclusion and Extensions

AOT reframes preference alignment as enforcing *distributional stochastic dominance* rather than pointwise margins. It reduces to a simple 1D OT problem solvable by sorting, yielding strong theoretical guarantees and excellent empirical performance. Recent work extends AOT to **multivariate stochastic dominance** via entropic regularization, enabling multidimensional alignment (helpfulness, safety, honesty) with improved correlation to human preference rankings.

References

[MMB+24] Igor Melnyk, Youssef Mroueh, Brian Belgodere, Mattia Rigotti, Apoorva Nitsure, Mikhail Yurochkin, Kristjan Greenewald, Jiri Navratil, and Jarret Ross. Distributional preference alignment of llms via optimal transport. *Advances in Neural Information Processing Systems*, 37:104412–104442, 2024.

[RSM+23] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems*, 36:53728–53741, 2023.